

HW1: Spot the Artifact

Teacher's Guide & Solution Key

Course: Machine Learning for Single-Cell and Spatial Biology (Prof. Uthsav Chitra, JHU)

Unit 1 — Linear Dimensionality Reduction (PCA, NMF, GLM-PCA) | Prepared by Sai Dharmasena

Overview

This homework gives students 5 simulated single-cell datasets. Each dataset contains a hidden technical artifact. Students must:

- Run PCA (and optionally NMF) on each dataset
- Identify the artifact by inspecting PCA plots, loadings, and correlations
- Propose and apply a correction
- Compare before/after and explain what happened

The datasets are designed so that **naive PCA fails to reveal the true biological structure** (3 cell types: TypeA, TypeB, TypeC), and students must figure out why.

Dataset specs: 500 cells × 200 genes. Each cell type has ~15 marker genes upregulated ~5-fold. All datasets share this clean biological backbone; the artifacts are layered on top.

Dataset 1: Scaling Artifact

Files: dataset1_scaling_expression.csv, dataset1_scaling_metadata.csv

The Artifact

Gene_100 is a "housekeeping" gene with very high constant expression (mean ~500 counts) but **no biological variation across cell types** (CV ~0.04). Its raw variance is ~400, while the median gene variance is ~2 — a 200x difference.

What Students Should Observe

- **Naïve PCA on raw counts:** PC1 is perfectly correlated with Gene_100 ($r = 0.99$). The 3 cell types are barely separated (silhouette = 0.31).
- **Key diagnostic:** Check the PC1 loadings — Gene_100 will have the largest loading by far. Check gene variances — Gene_100's variance dwarfs all others.
- **The trap:** Gene_100 looks "highly variable" by raw variance, but it's only variable because of its high mean, not because of biology.

The Fix

Log-normalize (library-size scaling + log1p) then **z-score each gene** (StandardScaler). This equalizes variance across genes regardless of mean. After correction: silhouette = 0.82, 3 cell types clearly separated.

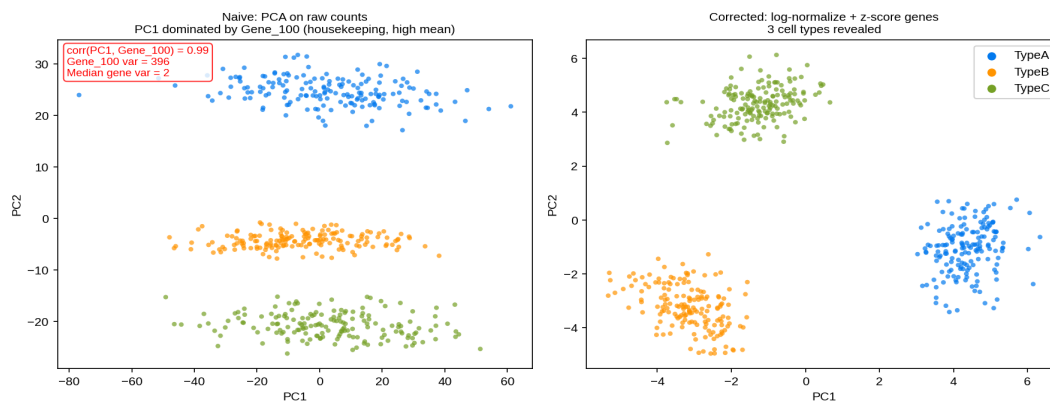


Figure 1: Naive PCA on raw counts (left) vs. corrected log-normalize + z-score (right).

Slide Connection

- **Class 03, slides 9–10:** "Informative: has large spread (variance)" — but variance depends on scale!
- **Class 03, slide 49:** Normalization challenges in PCA.

Answer Key

Metric	Naïve (raw PCA)	Corrected (log + z-score)
Silhouette	0.31	0.82
corr(PC1, Gene_100)	0.99	-0.16

Dataset 2: Zero-Fraction Artifact

Files: dataset2_zero_fraction_expression.csv, dataset2_zero_fraction_metadata.csv

The Artifact

Cells have highly variable library sizes (droplet capture efficiency varies 100x). Low-library-size cells have many more zeros. After a naive log1p transform (without library-size normalization), PC1 tracks the **fraction of zeros per cell** rather than cell type.

What Students Should Observe

- **Naive log1p + PCA:** PC1 is strongly correlated with zero fraction ($r = -0.89$). Cell types are mixed (silhouette = 0.07).
- **Key diagnostic:** Color cells by zero fraction (or library size) in PCA space. If PC1 aligns with zero fraction, that's the artifact.
- **The trap:** Even after log-transform, without proper library-size normalization, the zero fraction dominates.

The Fix

Pearson residuals (the class_05 approach): model counts as Poisson with gene-specific means and cell-specific size factors, compute residuals, then PCA. Standard log-normalization + z-score helps partially but doesn't fully remove the artifact (this is the key motivation for GLM-PCA). Pearson residuals fully correct: silhouette = 0.69, $\text{corr}(\text{PC1}, \text{zero_frac}) = -0.05$.

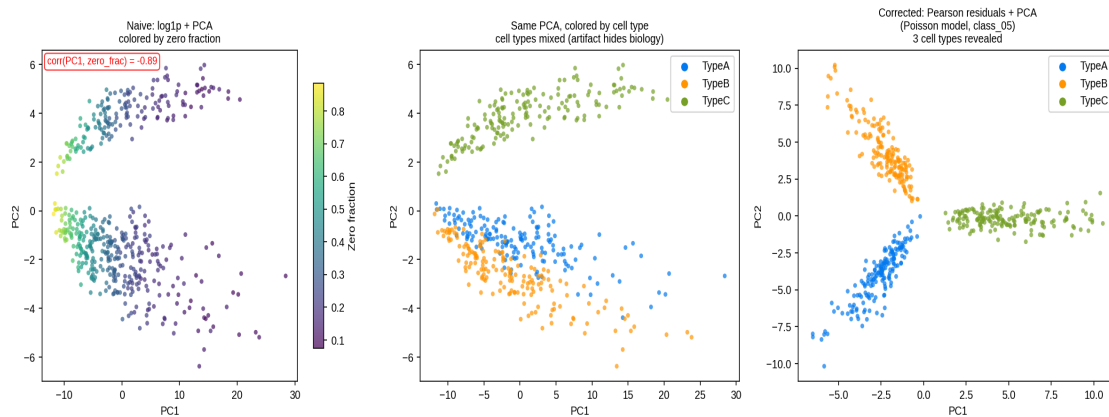


Figure 2: Naive log1p + PCA by zero fraction (left), by cell type (center), Pearson residuals (right).

Slide Connection

- **Class 05, slide 10:** "Top PC captures zero fraction" — this is the exact artifact.
- **Class 05, slides 8–9:** Normalization distorts counts; PCA assumes continuous data but UMI counts are discrete.
- **Class 05, slides 18–20:** GLM-PCA / Poisson-PCA as the fix.

Answer Key

Metric	Naive (log1p)	Standard (log+z)	Pearson residuals
Silhouette	0.07	0.14	0.69
$\text{corr}(\text{PC1}, \text{zero_frac})$	-0.89	-0.89	-0.05

Pedagogical note: The standard pipeline (log-norm + z-score) does NOT fix this artifact well. This is intentional — it motivates why GLM-PCA / deviance residuals are needed (the central message of class_05).

Dataset 3: Outlier Artifact

Files: dataset3_outliers_expression.csv, dataset3_outliers_metadata.csv

The Artifact

10% of cells are outliers (simulating doublets or damaged cells). Each outlier has aberrantly high expression in a random subset of 10–25 genes (200 extra counts). These outliers hijack the top PCs.

What Students Should Observe

- **Naïve PCA (with outliers):** A cluster of outlier cells separates from the main cloud along PC1. The 3 cell types within the main cloud are partially obscured.
- **Key diagnostic:** Look for a small, distinct cluster far from the main cloud. Check total counts per cell — outliers will have 2x higher totals. Check for cells with extreme expression in genes that aren't markers.
- **The trap:** Outliers inflate variance in random genes, pushing biological signal out of the top PCs.

The Fix

Remove outliers (e.g., filter cells with total counts > 2 standard deviations above mean, or use a doublet-detection approach), then re-run PCA. After removal: silhouette = 0.80, 3 cell types clearly separated.

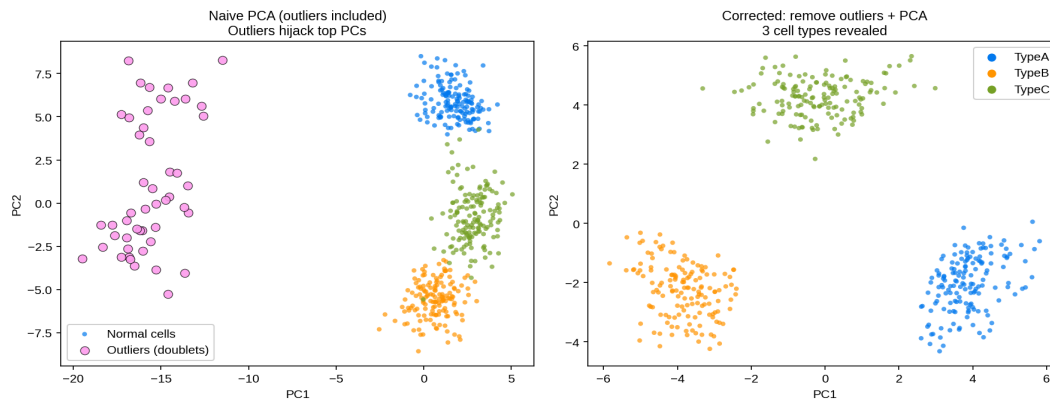


Figure 3: Naïve PCA with outliers (left) vs. corrected after outlier removal (right).

Slide Connection

- **Class 03, slide 46:** Batch effects and technical variation in PCA.
- **Class 01, slide 24:** Technical challenges in single-cell data.
- General QC theme: always inspect for outliers before dimensionality reduction.

Answer Key

Metric	Naïve (with outliers)	Corrected (outliers removed)
Silhouette (normal cells)	0.63	0.80
Outliers	50 cells (10%)	0

Dataset 4: Batch Effect Confounding

Files: dataset4_batch_expression.csv, dataset4_batch_metadata.csv

The Artifact

Cells come from 2 batches (Batch1, Batch2), randomly assigned (not confounded with cell type). Batch2 has global 1.5x upregulation plus 2x upregulation of genes 150–200. The batch effect is stronger than the cell-type signal, so PC1 separates batches instead of cell types.

What Students Should Observe

- **PCA colored by batch:** PC1 cleanly separates Batch1 from Batch2 (batch silhouette = 0.48).
- **PCA colored by cell type:** Cell types are mixed — the batch effect dominates (type silhouette = 0.26).
- **Key diagnostic:** Color the same PCA plot by batch AND by cell type. If batch separates along PC1 but cell types don't, batch effect is confounding.
- **The trap:** PC1 looks "informative" (high variance explained) but it's capturing technical variation, not biology.

The Fix

Batch correction: center each batch separately (subtract per-batch mean), then PCA. More sophisticated methods (Harmony, CCA/Seurat anchors) would also work. After correction: type silhouette = 0.76, batch silhouette = 0.00.

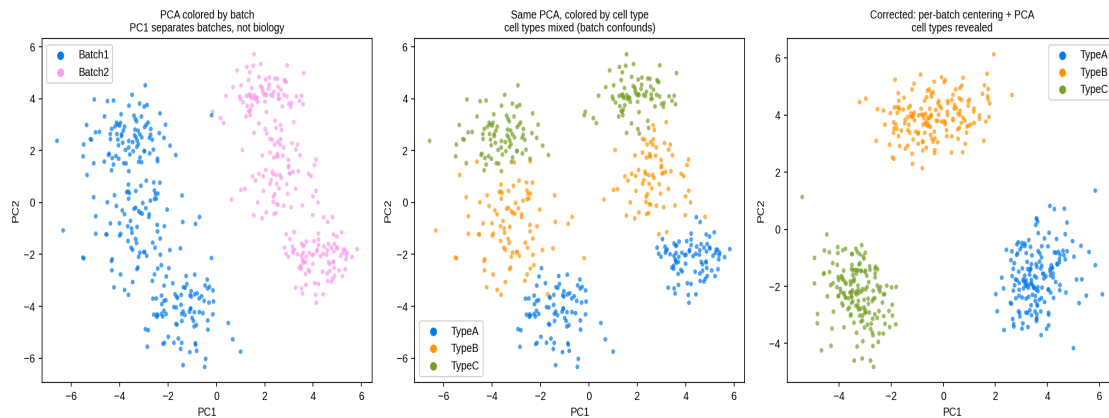


Figure 4: PCA by batch (left), by cell type (center), per-batch centered (right).

Slide Connection

- **Class 03, slide 46:** "Different individuals are separated in PC space (batch effect)."
- **Class 05, slides 28–29:** CCA and Seurat anchor-based integration for batch correction.
- **Class 04, slides 36–38:** LIGER for data integration (shared + dataset-specific factors).

Answer Key

Metric	Before correction	After correction
Cell type silhouette	0.26	0.76
Batch silhouette	0.48	0.00

Dataset 5: Nonlinear Manifold

Files: dataset5_nonlinear_expression.csv, dataset5_nonlinear_metadata.csv

The Artifact

Cells lie on a curved 2D manifold (a Swiss-roll-like structure parameterized by pseudotime). The true structure is nonlinear — cells progress continuously from Early to Late stage along a curve. PCA, being linear, flattens this curve and cannot capture the continuous trajectory.

What Students Should Observe

- **PCA:** The 3 stages (Early, Mid, Late) appear partially separated but the continuous, curved structure is lost. PCA projects the swiss roll onto a flat plane.
- **Key diagnostic:** Color by pseudotime (continuous). If the colors form a gradient that bends or curves in a way PCA can't capture, the structure is nonlinear.
- **The trap:** PCA assumes linear structure. If the data lies on a curved manifold, PCA gives a misleading flattened projection.

The Fix

PCA cannot fix this — it's a linear method. This motivates nonlinear methods: t-SNE, UMAP, diffusion maps, PHATE, etc. (covered in Unit 2: graphs and manifold learning). Students should recognize that the "fix" is to use a different method, not to preprocess differently.

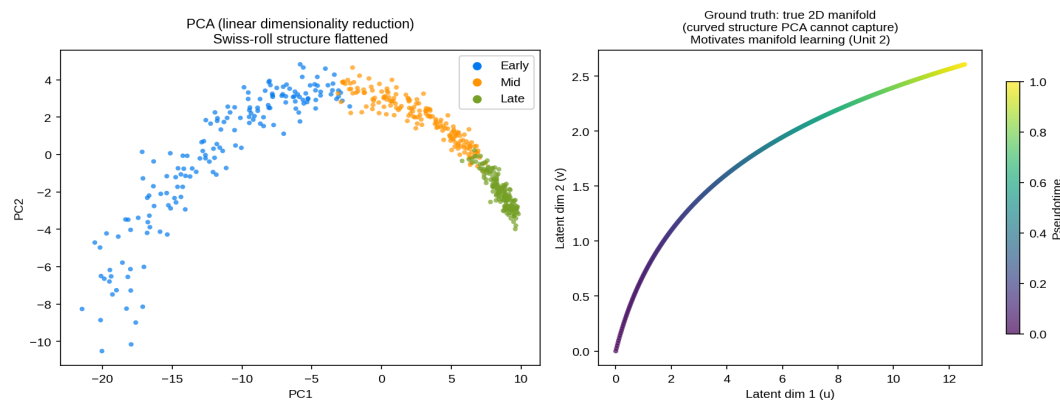


Figure 5: PCA flattens the Swiss-roll (left). Ground truth manifold (right) shows the true curved trajectory.

Slide Connection

- **Class 01, slide 31:** "Part 2: graphs and manifold learning — neighborhood graphs, random walks, diffusion, clustering, trajectories."
- **Class 03, slide 20:** PCA finds linear directions of max variance — but what if structure is nonlinear?
- This dataset is a teaser for Unit 2.

Answer Key

Metric	PCA (linear)	Ground truth
Silhouette by stage	0.52	N/A (continuous)
Structure	Flattened	Curved (Swiss roll)

Suggested Homework Structure, Rubric & Notes

Deliverables (per dataset) & Optional Extensions

- **PCA plot** (PC1 vs PC2) with chosen coloring
- **Diagnosis:** What artifact? What evidence?
- **Correction:** What did they do to fix it?
- **Comparison plot:** Before vs after
- **Brief reflection:** 2–3 sentences
- Run NMF on datasets 1–4; compare to PCA
- Implement GLM-PCA for dataset 2; compare to standard PCA
- Try UMAP/t-SNE on dataset 5; show it recovers the curve

Grading Rubric (per dataset)

Criterion	Points
Correctly identifies the artifact	3
Provides evidence (plot, correlation, loading analysis)	3
Applies appropriate correction	2
Compares before/after with plot	1
Reflection demonstrates understanding	1
Total per dataset / Total (5 datasets)	10 / 50

File Inventory

hw1_artifact_datasets/ data/ (10 CSVs: expression + metadata per dataset) figures/ (5 PNG/SVG: fig1-fig5)
teacher_guide/HW1_Teacher_Guide.pdf

data/ dataset1_scaling_{expression,metadata}.csv | dataset2_zero_fraction_{expression,metadata}.csv |
dataset3_outliers_{expression,metadata}.csv

dataset4_batch_{expression,metadata}.csv | dataset5_nonlinear_{expression,metadata}.csv

figures/ fig1_scaling | fig2_zero_fraction | fig3_outlier | fig4_batch | fig5_nonlinear (.png/.svg)

Other Notes

- **Difficulty progression:** Datasets 1–3 are "spot and fix" (preprocessing). Dataset 4 requires recognizing confounding. Dataset 5 is conceptual (PCA's limitation, motivates Unit 2).
- **Dataset 2 is the keystone:** It directly mirrors class_05 slide 10 and motivates GLM-PCA. The fact that standard normalization doesn't fully fix it is the pedagogical punchline.
- **All datasets are small (500 x 200)** so students can run everything quickly, even with an agent.
- **Metadata files include ground-truth labels** (cell_type, batch, is_outlier, pseudotime) — students should NOT use these for their analysis, only for verifying their findings. Consider withholding metadata and having students discover structure first.
- **Random seed = 42** for reproducibility. Generation code is in the notebook.